TEAM SEAT 02 OF 04 · INSTITUTIONAL QUANTITATIVE DESK RESEARCH MANUAL

QUANTITATIVE & MATHEMATICAL RESEARCHER SPECIFICATION

Role Mandate: Econometric Regime Proofs, Stochastic Processes, Backtest Rigor & MLOps System: trading-sandbox v2.4

YOUR MISSION INSIDE THE 4-PERSON QUANTITATIVE TEAM

As the **Quantitative & Mathematical Researcher (Seat 02)**, you are the scientific engine of the desk. Operating from the local Research Desk (`stats_analyzer.py` , `strategies/` , `ParameterGridSearch.py` , and `ai/`), your job is to formulate falsifiable market hypotheses, prove time-series stationarity, eliminate look-ahead and data-snooping biases, and train Purged Cross-Validation Meta-Labeling models (`models/rf_meta_model.pkl`) that pass the **Finance Officer's (Seat 01)** Sharpe and friction gates before the **Software Engineer (Seat 03)** deploys them to Google Cloud.

2.1 ECONOMETRIC REGIME DETECTION: STATIONARITY & MEMORY PROOFS

Never design an entry rule before proving the econometric regime of the price series P_t or synthetic spread z_t. In `RegimeAnalyzer` (`stats_analyzer.py`), every asset is tested via the **Augmented Dickey-Fuller (ADF)** unit-root regression and the rescaled-variance **Hurst Exponent (H)** across lags $k \in [2, 20)$:

ADF Unit-Root Test: $\Delta y_t = \alpha + \beta t + \gamma y_{t-1} + \sum_{i=1..p} \delta_i \Delta y_{t-i} + \varepsilon_t \quad (H_0: \gamma = 0)$

Hurst Variance Scaling: $\tau(k) = \sqrt{(Std(y_{t+k} - y_t))} \propto k^H \Rightarrow polyfit(ln(k), ln(\tau(k)), 1)$

Statistic	Empirical Threshold	Regime State	Mathematical Interpretation & Strategy Routing
Raw Price ADF *p*	$p > 0.05$ (SPY: *0.7411*)	Non-Stationary Level	Fails to reject H_0 ($\gamma = 0$); raw prices drift unboundedly. Requires log-differencing $r_t = ln(P_t / P_{t-1})$ or cointegrated spread construction.
Log Return ADF *p*	$p < 0.05$ (SPY: *0.0000*)	Stationary Increments	Rejects unit root; confirms return distribution is covariance-stationary and valid for rolling volatility and Z-score estimation.
Hurst $H < 0.48$	Anti-Persistent (SPY: *0.4647*)	**MEAN_REVERTING**	Variance grows slower than $k^{0.5}$. Route to `BollingerMeanReversion` (`strategies/mean_reversion.py`) or Cointegrated Pairs.
Hurst $0.48 \le H \le 0.52$	Geometric Brownian Motion	**RANDOM_WALK**	Per Malkiel's EMH, autocorrelation is zero and transaction friction exceeds alpha. System forces `signal = 0` (100% Cash).
Hurst $H > 0.52$	Persistent Memory	**TRENDING**	Positive autocorrelation driven by slow information diffusion (PEAD) or institutional flow. Route to Dual SMA / Breakout modules.

2.2 COINTEGRATION VS. CORRELATION & ORNSTEIN-UHLENBECK HALF-LIFE

In multi-asset statistical arbitrage, **correlation does not imply cointegration**. For example, Coca-Cola (`KO`) and Pepsi (`PEP`) exhibit a significant daily return correlation of $\rho = 0.4849$, yet their price spread diverges permanently over decades. Conversely, gold ETF (`GLD`) and gold miners ETF (`GDX`) pass the Engle-Granger two-step CADF test, forming a stationary spread $z_t = y_t - \beta x_t$. Rather than P-hacking lookback windows, model $z(t)$ as a continuous-time **Ornstein-Uhlenbeck (OU) process** to derive the exact analytical holding period (e.g., ~10 days for `GLD/GDX`):

Ornstein-Uhlenbeck SDE: $dz(t) = -\theta (z(t) - \mu) dt + \sigma dW_t \Rightarrow$ **Analytical Half-Life:** $t_{1/2} = ln(2) / \theta$ |
Bollinger Z-Score: $Z_t = (P_t - \mu_{20}) / \sigma_{20}$

2.3 MLOPS ARCHITECTURE: TRIPLE-BARRIER, META-LABELING & PURGED CV

Training neural networks to predict raw prices fails due to low signal-to-noise ratios. Our `ai/` architecture implements Marcos López de Prado's institutional framework: the primary econometric strategy determines trade direction (*signal* $\in \{-1, +1\}$), while a secondary Random Forest / XGBoost classifier (`ai/meta_labeler.py`) predicts the probability $P(Y = 1)$ that the trade hits its upper profit barrier before its stop-loss or vertical time expiration (`ai/triple_barrier.py`).

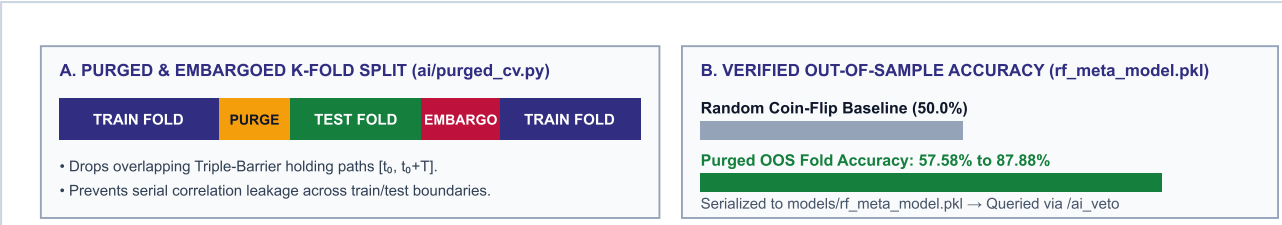


Figure 2.1: Purged K-Fold Cross-Validation geometry (`ai/purged_cv.py`) and verified Out-of-Sample accuracy (57.58%–87.88%) of our trained Random Forest Meta-Labeler.

2.4 MANDATORY BACKTEST HYGIENE & ANTI-OVERFITTING CHECKLIST

Every script in `strategies/` and `ParameterGridSearch.py` must obey five mathematical laws before presentation to Seat 01:

1. **1-Bar Forward Execution Shift (Look-Ahead Prevention):** Signals computed from bar t close must execute at $t + 1$ (`data["position"] = data["signal"].shift(1)`). Verify by truncating the last N days of data and confirming prior positions remain identical.
2. **Parameter Parsimony (Data-Snooping Prevention):** Restrict free parameters to ≤ 5 and require at least $252 \times N_{params}$ daily observations. Reject isolated parameter spikes in `ParameterGridSearch.py`; select broad stability plateaus.
3. **Split/Dividend & Survivorship Hygiene:** Use split- and dividend-adjusted Close prices (multiplying prior prices by $(Close_{T-1} - d) / Close_{T-1}$) rather than noisy intraday High/Low ticks, and account for point-in-time index survivorship.
4. **Automated Research Ingestion:** Use Discord `#research-agent` (`!arxiv` powered by Groq `qwen/qwen3.8-27b`) and Web UI Tab 2 (*Ernest Chan Strategy Hub* — SSRN, NBER, arXiv `q-fin`) to source new academic factor models.

2.5 CROSS-TEAM HANDOFF PROTOCOL: HOW SEAT 02 WORKS WITH SEATS 01, 03 & 04

Team Counterpart	Your Deliverable to Them	What You Require Back From Them
Seat 01: Finance & Risk Officer	Markdown grid-search tables (`#backtests`) showing Net Return (after 8 bps friction), Sharpe Ratio, Max Drawdown, and OOS Meta-Label accuracy.	Written approval to promote the parameter set to Paper/Live trading and Half-Kelly covariance weight allocation.
Seat 03: Software Engineer	Decoupled strategy class inside `strategies/*.py` implementing `generate_signals(df) -> pd.DataFrame` plus serialized `models/*.pkl`.	Clean Parquet caches in `data/` (`data_fetcher.py`), unit test coverage in `tests/`, and deployment to the Google Cloud VM.
Seat 04: General Operations	Updated strategy cards in Web UI Tab 2 and plain-English `/ai_veto` probability thresholds.	Feedback on live signal frequency and `!arxiv` research digest curations.

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